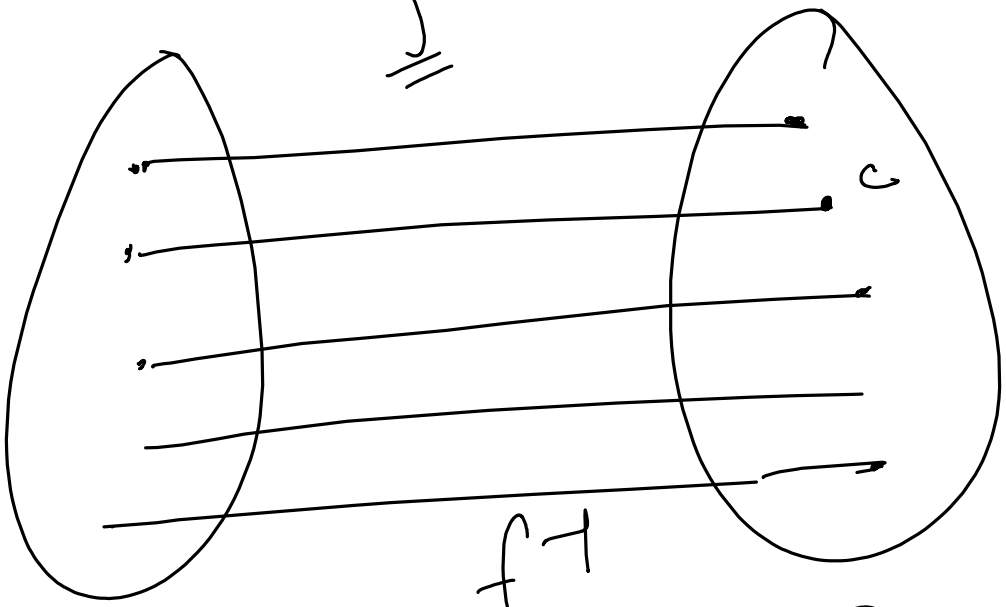
ONE TO ONE

f



Domain f^{-1} Range

f one to one & onto

$m \rightarrow H(m)$

If $m_1 \neq m_2$

$f(m_1) \neq f(m_2)$

ONTO



$f(D) = R$

bijection

$\exists\, m_1, m_2$

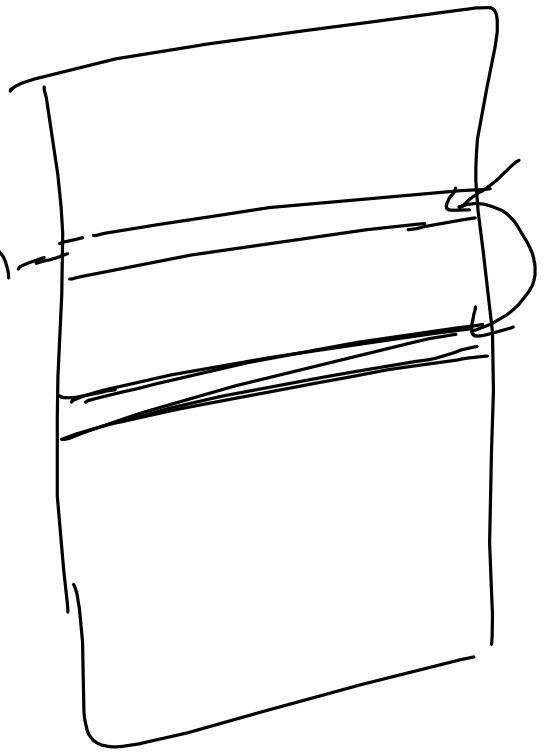
$H(m_1) = H(m_2)$

Hash in data Structure

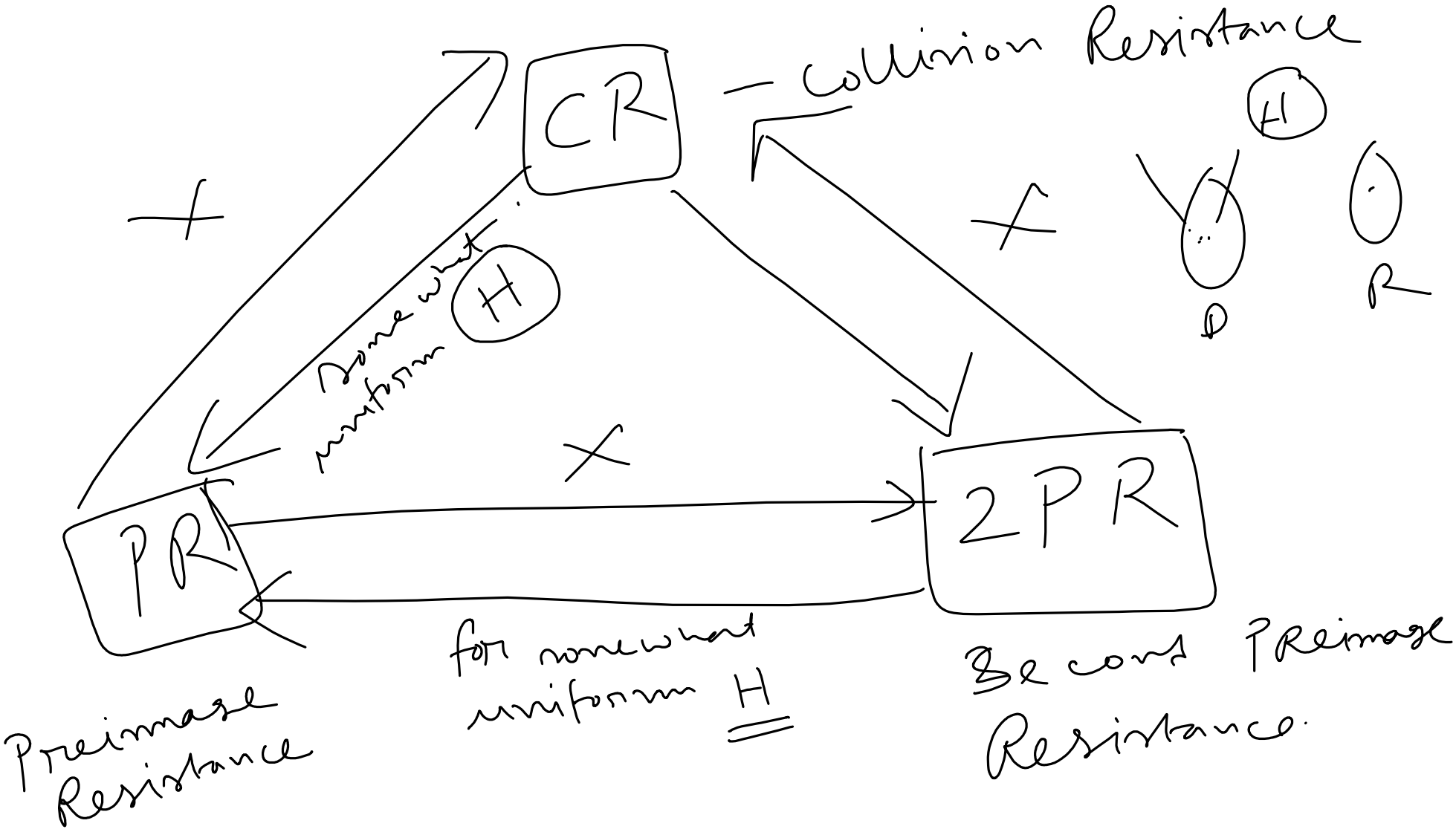
x f(x)

CRYPTOGRAPHY

collisions are not easily detectable by a PPT (Probabilistic polynomial algo)

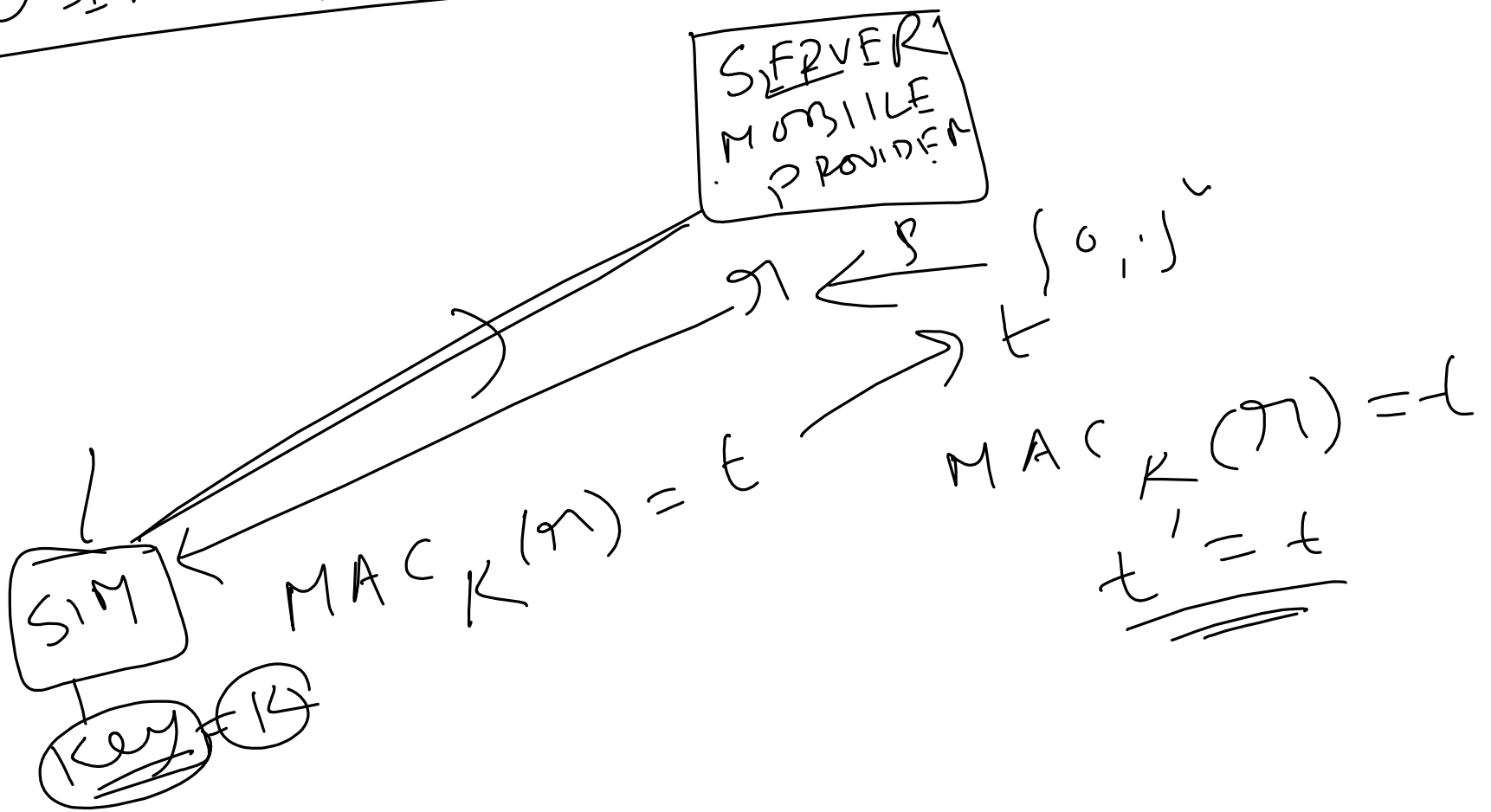


collision resolution strategy


CR
Collision Resistance
H
Some what uniform
H
PR
2PR
P
R
Preimage Resistance
for somewhat uniform H
Second Preimage Resistance.

USER AUTHENTICATION

~~MAC~~ MAC



CANNONICAL VERIFICATION

Alice (K) — Bob (K)

(m, t) → (m, t)

(t) $Mac_K(m)$

Verify (m, t) = 1 / 0

deterministic function

Bob calculates $t' = Mac_K(m)$

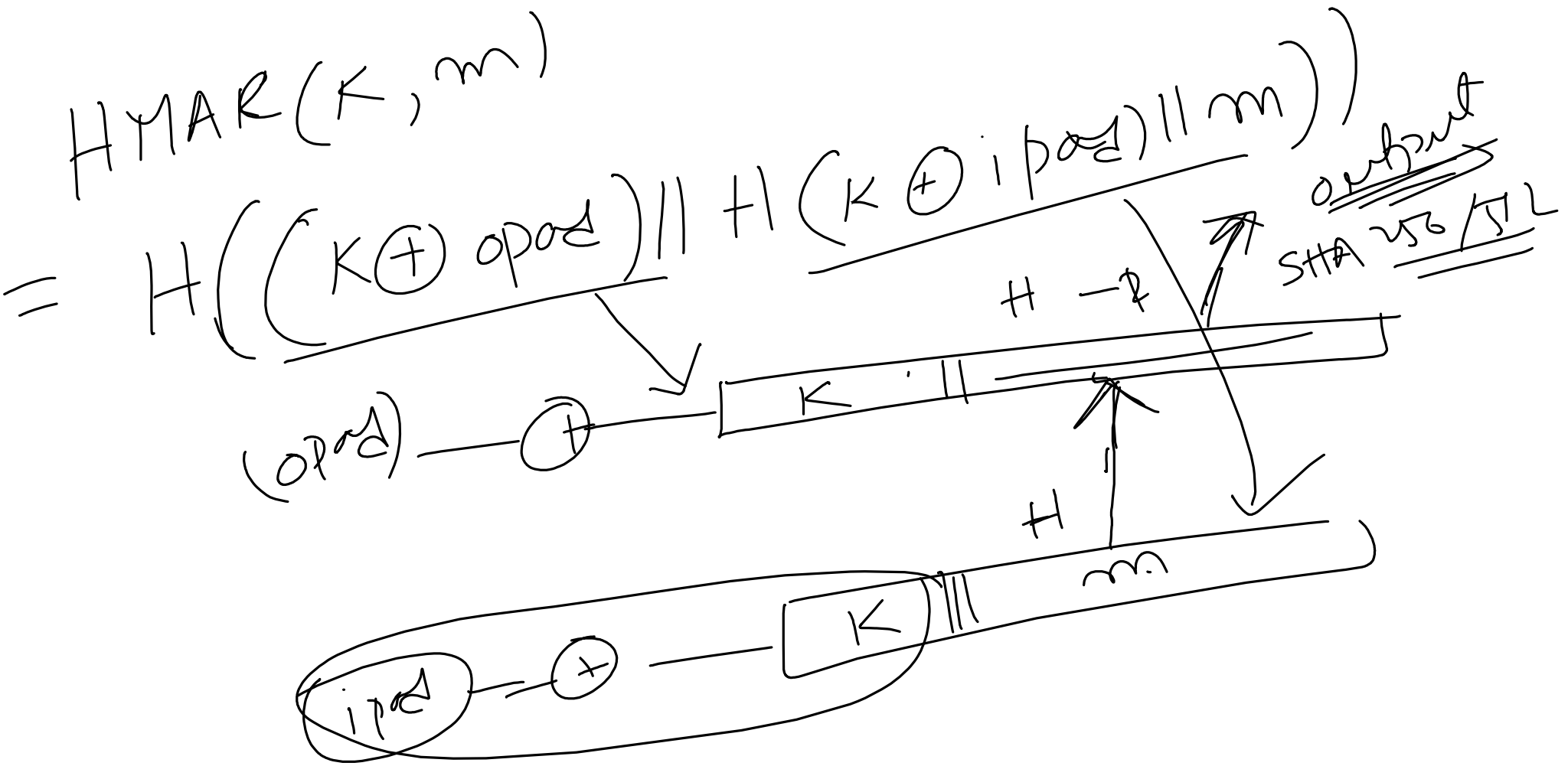
is $t = t'$?

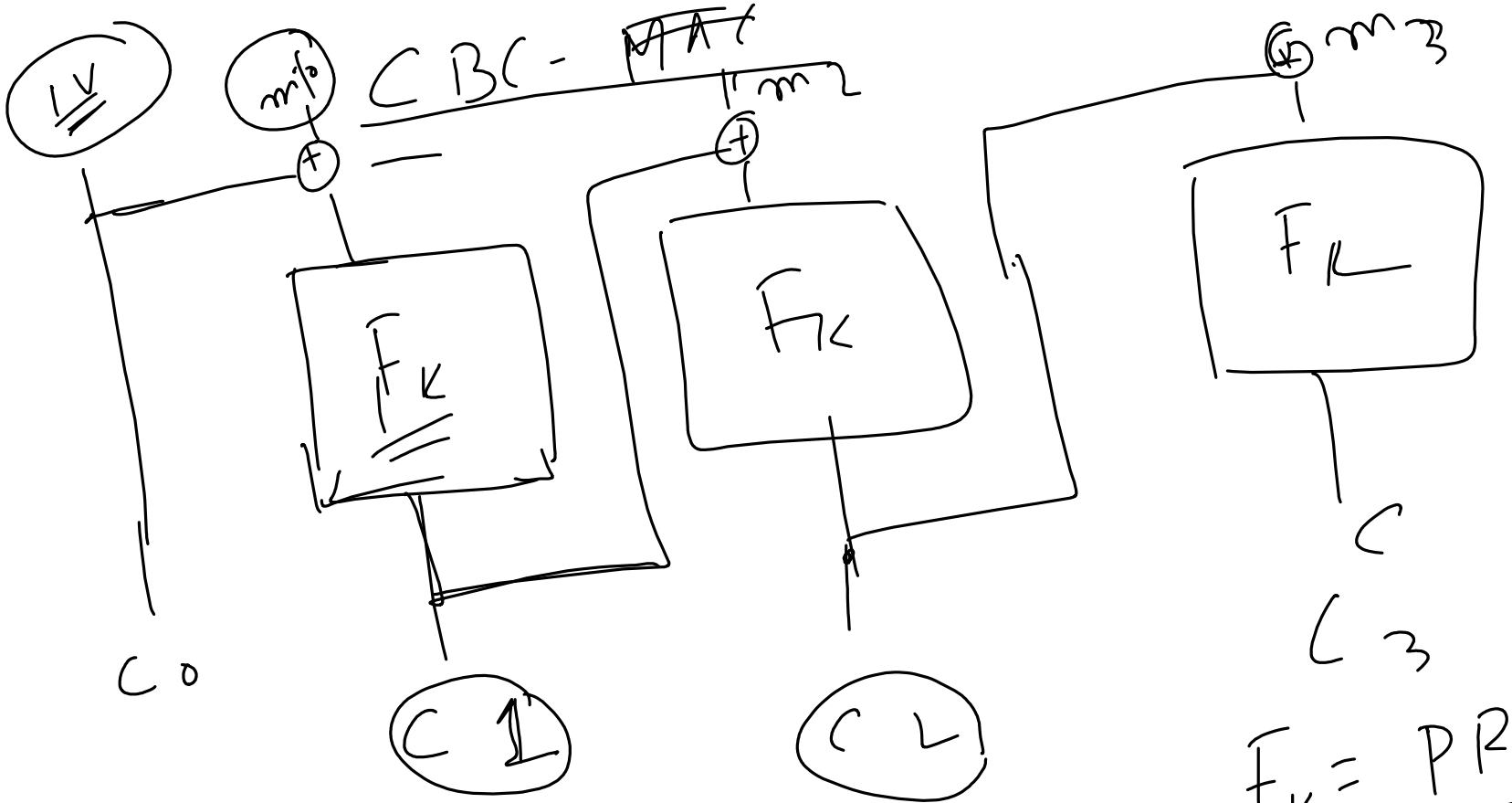
if yes then verify succeeds

HMAC

HMAC(K, m)

$= H((K \oplus opad) \| H((K \oplus ipad) \| m))$





$m_1 m_2 m_3$

3b - 4 bits

4 bits possible

256

F_K = PRF

AES

CBC

$m, (IV, t)$ $t = c_3$

non-repudiation

Alice

BB

$r \xleftarrow{\$}$ {0,1}^λ

m

$H(m \| r) = t$

Hiding $\leftarrow$ $m \| r$

m

r

BB

$H(m \| r) = t'$

$t = t'$

m', r' ✓

Binding $\leftarrow$ $H(m \| r) = H(m' \| r')$

$m = m'$

Commitment scheme